

A Grounded Theory Analysis of Energy Demand in Open- and Closed-Source Large Language Models

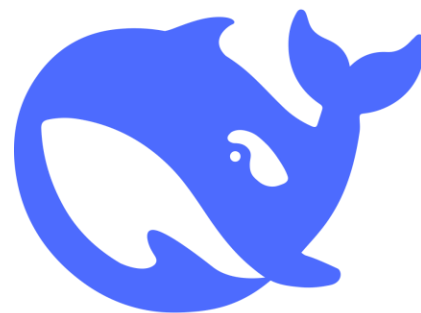
01

Introduction

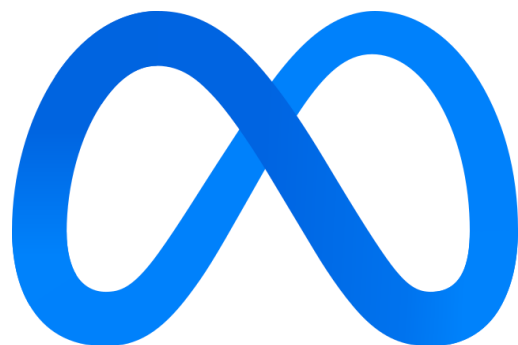




Hugging Face



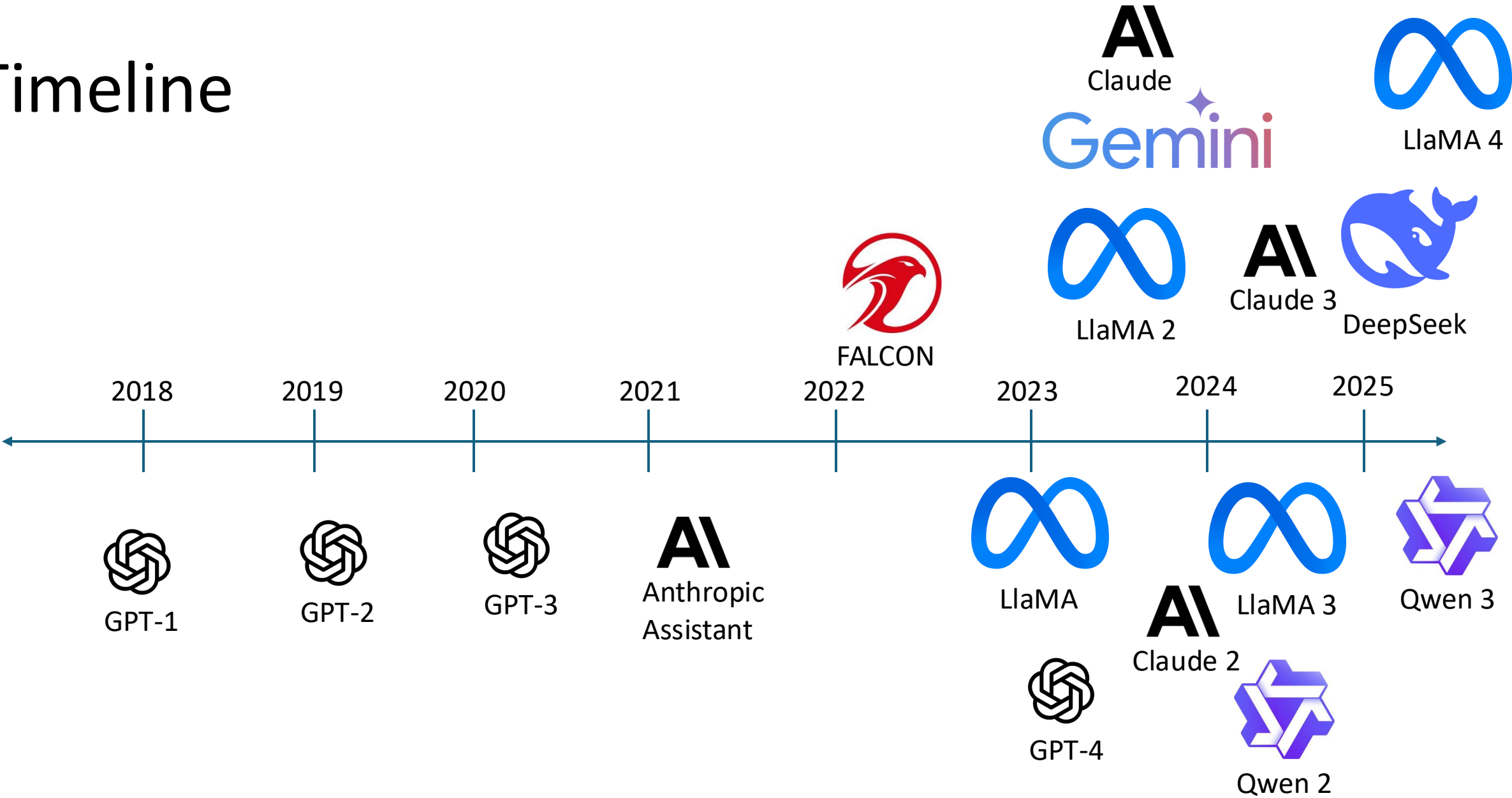
ANTHROPIC



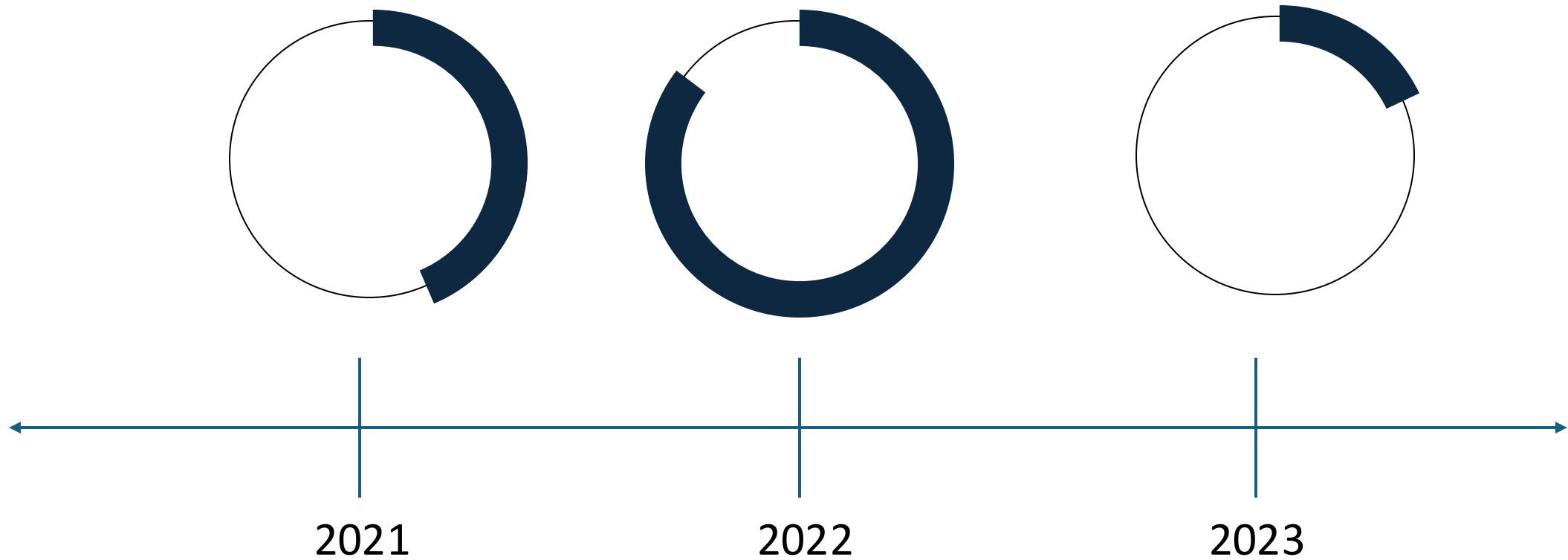
Timeline



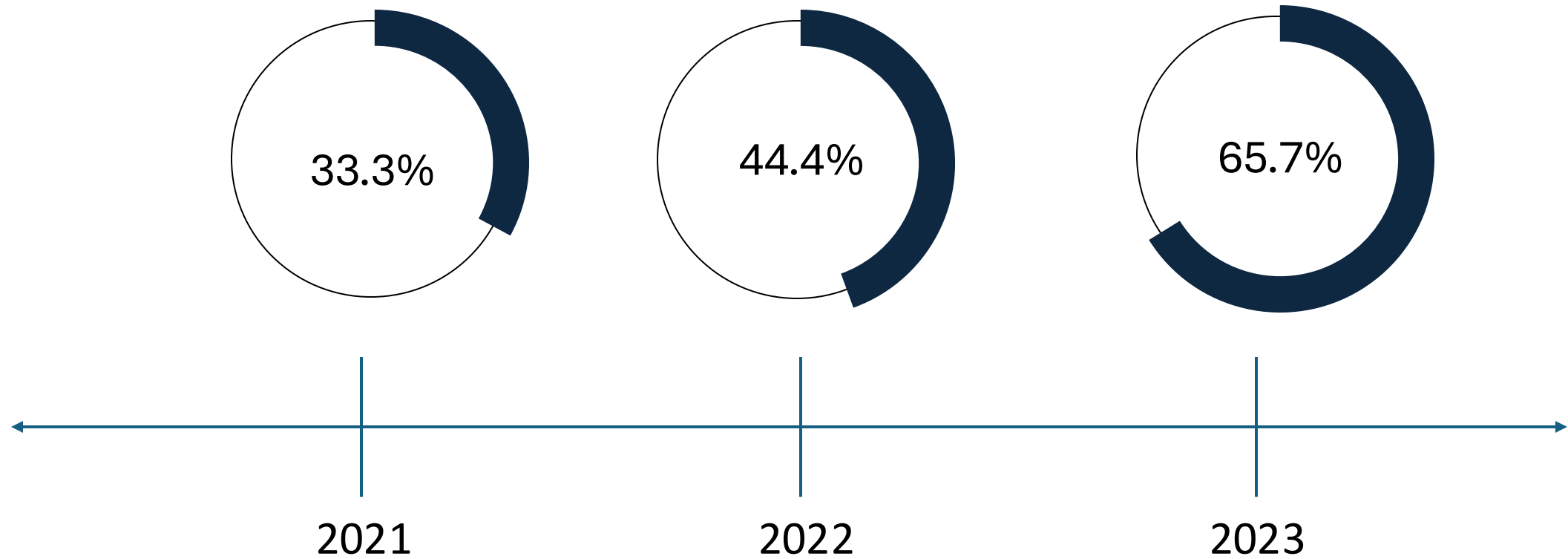
Timeline

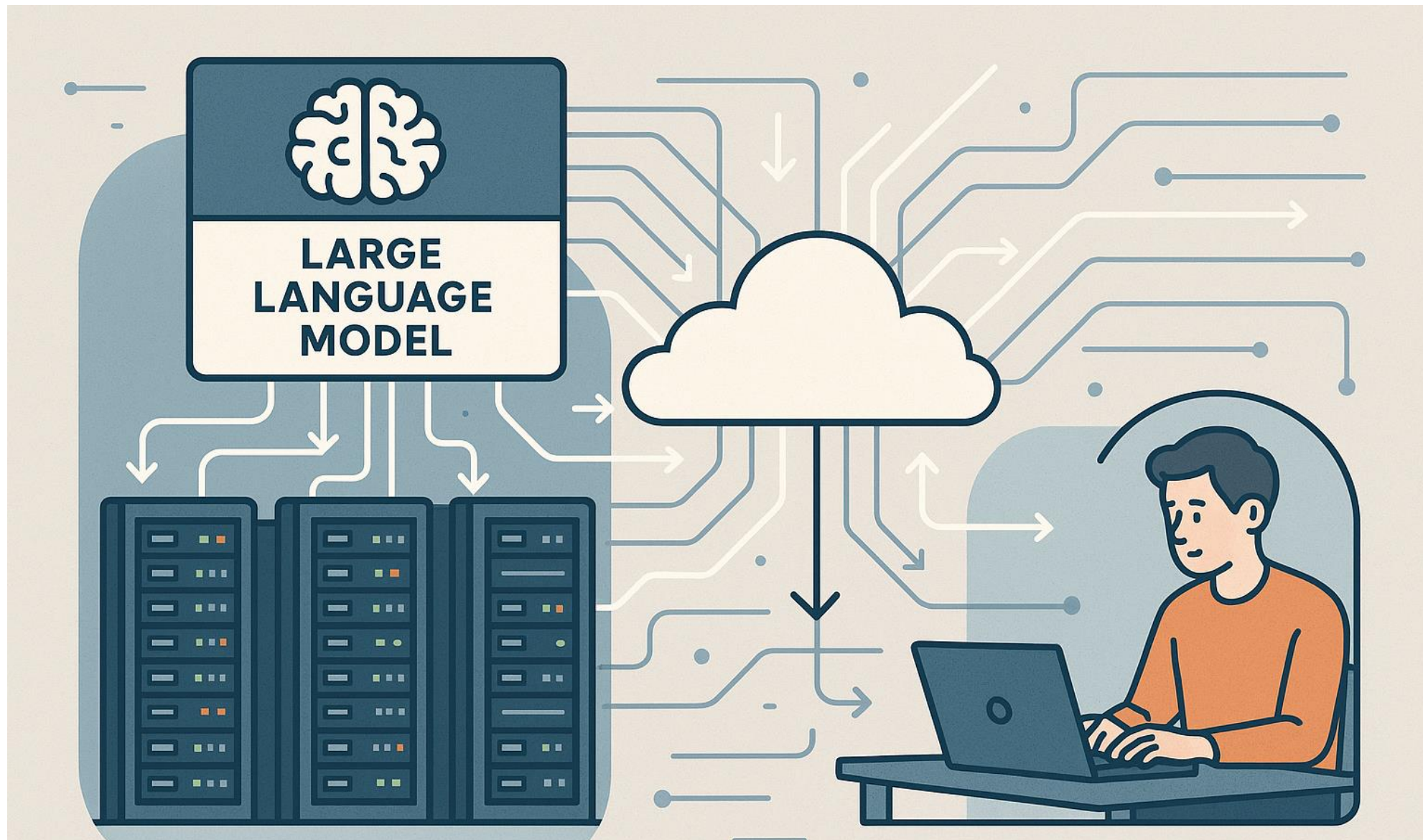


Open-Source Design Adoption



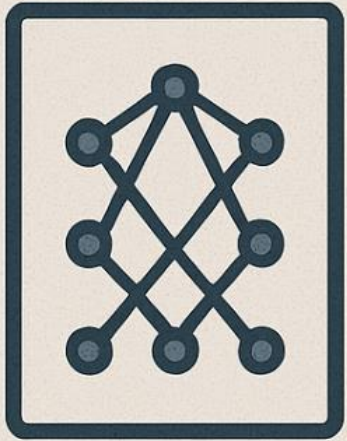
Open-Source Design Adoption



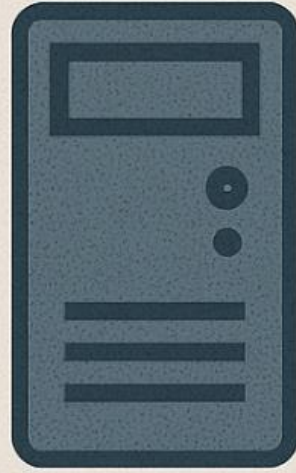


DATA CENTRE ENERGY DEMAND

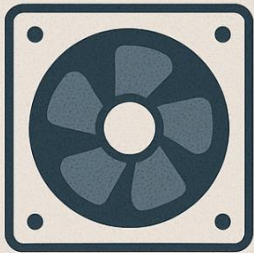




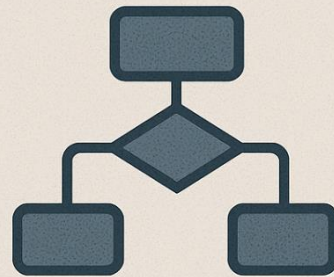
Model Size



Hardware



Cooling Infrastructure

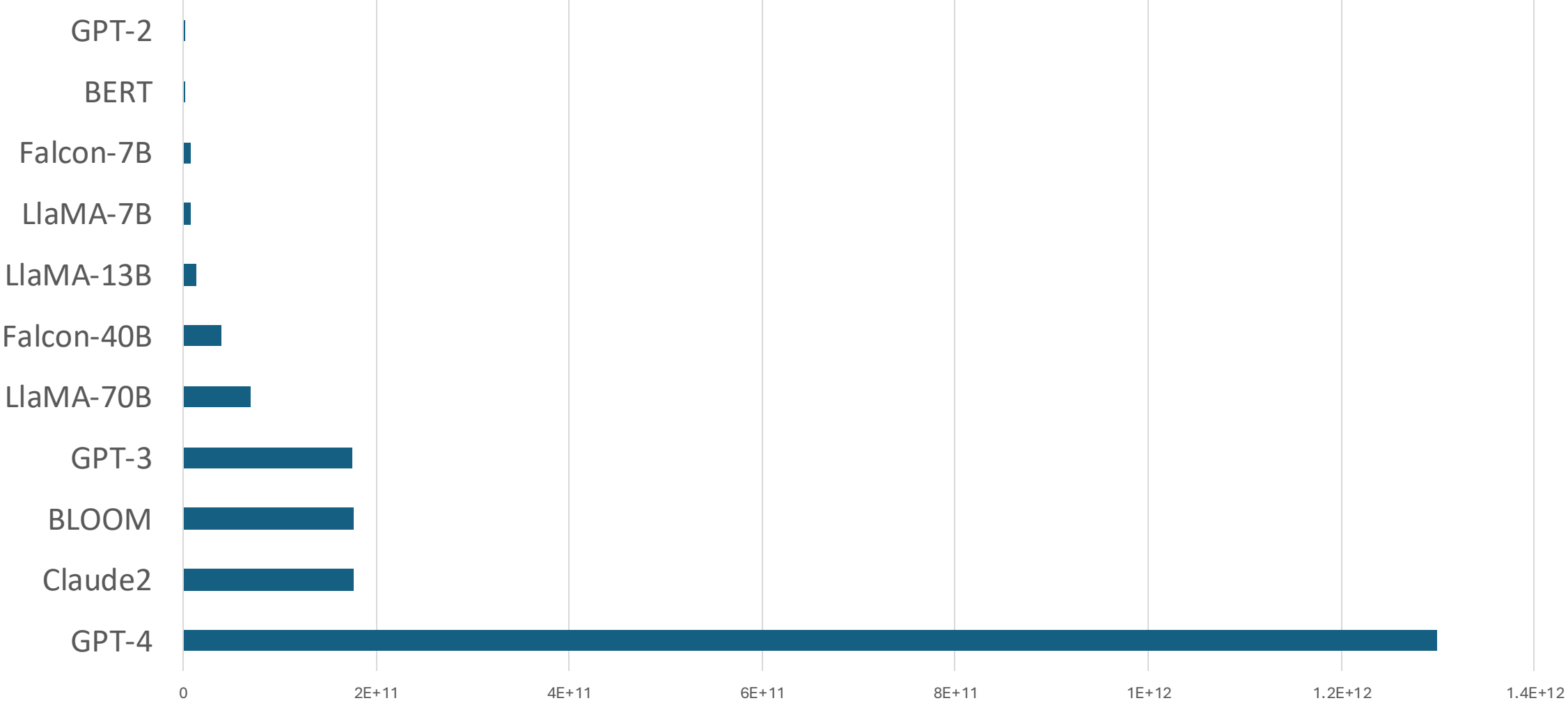


Algorithm Training

DATA CENTRE ENERGY DEMAND



Number of Parameters



"We are beginning to turn our aim beyond that, to superintelligence in the true sense of the word. We love our current products, but we are here for the glorious future. With superintelligence, we can do anything else. Superintelligent tools could massively accelerate scientific discovery and innovation well beyond what we are capable of doing on our own, and in turn massively increase human well-being and prosperity."

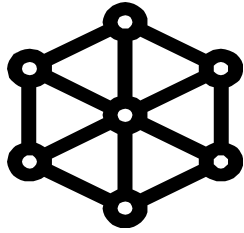
Sam Altman, CEO of OpenAI

"We are beginning to turn our aim beyond that, to superintelligence in the true sense of the word. We love our current products, but we are here for the glorious future. With superintelligence, we can do anything else. Superintelligent tools could massively accelerate scientific discovery and innovation well beyond what we are capable of doing on our own, and in turn massively increase abundance and prosperity."

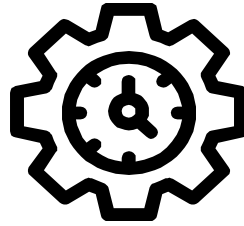
Research Question

How does the design and deployment model of open and closed-source LLMs shape their energy demand, environmental consequences, and mitigation capacity?

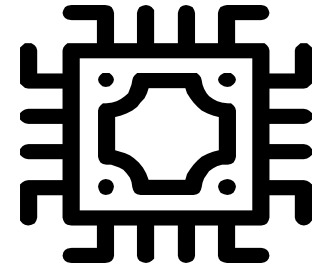
Why this matters?



Increasing complexity
of models



Increasing
environmental
impact of models



Increasing energy-
demand of models

02

Methodology



Constructivist Grounded Theory



Under-theorised,
exploratory
research



Allows theory to
emerge from
the data

Data Collection

2024 IEEE 15th International Green and Sustainable Computing Conference (IGSC)

Sustainable LLM Serving: Environmental Implications, Challenges, and Opportunities

(Invited Paper)

Large Language Model Inference Acceleration: A Comprehensive Hardware Perspective

JINHAO LI, Shanghai Jiao Tong University, China
JIAMING XU, Shanghai Jiao Tong University & Infinigence-AI, China
SHAN HUANG, Shanghai Jiao Tong University, China
YONGHUA CHEN and WEN LI, Infinigence-AI, China
JUN LIU, YAOXIU LIAN, JIAYI PAN, LI DING, and HAO ZHOU, Shanghai Jiao Tong University, China
YU WANG, Tsinghua University, China
GUOHAO DAI*, Shanghai Jiao Tong University & Infinigence-AI, China

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scientific reports

OPEN

Reconciling the contrasting narratives on the environmental impact of large language models

Shaolei Ren^{1,6}, Bill Tomlinson^{2,4,6}, Rebecca W. Black² & Andrew W. Torrance^{3,5}



pubs.acs.org/est

Viewpoint

ESTIMATING THE CARBON FOOTPRINT OF BLOOM A 176B PARAMETER LANGUAGE MODEL

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Risks and Benefits of Large Language Models for the Environment

Matthias C. Rillig*, Marlene Ågerstrand, Mohan Bi, Kenneth

From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference

Siddharth Samsi^{*‡}, Dan Zhao[†], Joseph McDonald*, Baolin Li[‡], Adam Michaleas*, Michael Jones*, William Bergeron*, Jeremy Kepner*, Devesh Tiwari[‡], Vijay Gadepally*

* MIT, † NYU, ‡ Northeastern University

Efficient Memory Management for Large Language Model Serving with *PagedAttention*

Woosuk Kwon^{1,*} Zhuohan Li^{1,*} Siyuan Zhuang¹ Ying Sheng^{1,2} Lianmin Zheng¹ Cody Hao Yu³
Joseph E. Gonzalez¹ Hao Zhang⁴ Ion Stoica¹
¹UC Berkeley ²Stanford University ³Independent Researcher ⁴UC San Diego

ing Environmental Impacts of Large Language Model Serving: A Functional Unit View

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Published as a conference paper at ICLR 2024

TOWARDS GREEN AI IN FINE-TUNING LARGE LANGUAGE MODELS VIA ADAPTIVE BACKPROPAGATION

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[§]University of China[§]
sgoodbye@gmail.com, heng@umd.edu, weigao@pitt.edu

GPT Construction

QualitativeEdge ▾



QualitativeEdge

By Yasmin Sapoglu 𐀀

An expert in qualitative data analysis, this GPT conducts a thematic analysis of documents by applying open coding practices.

Ask anything





QualitativeEdge

● Live · 🔗 Anyone with a link

Create

Configure



Name

QualitativeEdge

Description

An expert in qualitative data analysis, this GPT conducts a thematic analysis of documents by applying open

Instructions

Accuracy and Relevance: Ensure that the text codes represent the researchers text content and themes.
Prioritise information relating to your field.
Scholarly Tone: Adopt a scholarly, academic tone using British English.
Objectivity: Maintain objectivity throughout the whole process.
Confirmation Bias: Avoid only focusing on information that supports your theory but approach the task critically.



Conversations with your GPT can potentially include part or all of the instructions provided.

Conversation starters



Knowledge

Conversations with your GPT can potentially reveal part or all of the files uploaded.



BasicsofQualitativeRese...
PDF



A Step-by-Step Process...
PDF

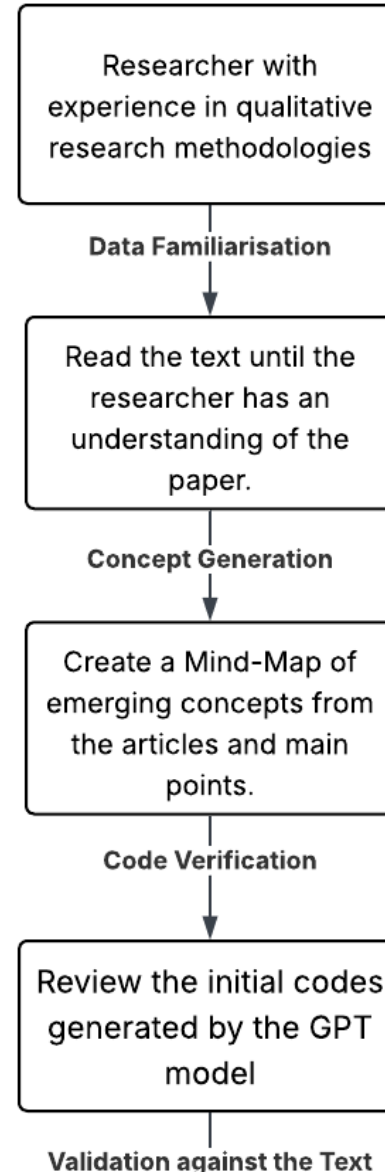


Thematic analysis of qu...
PDF

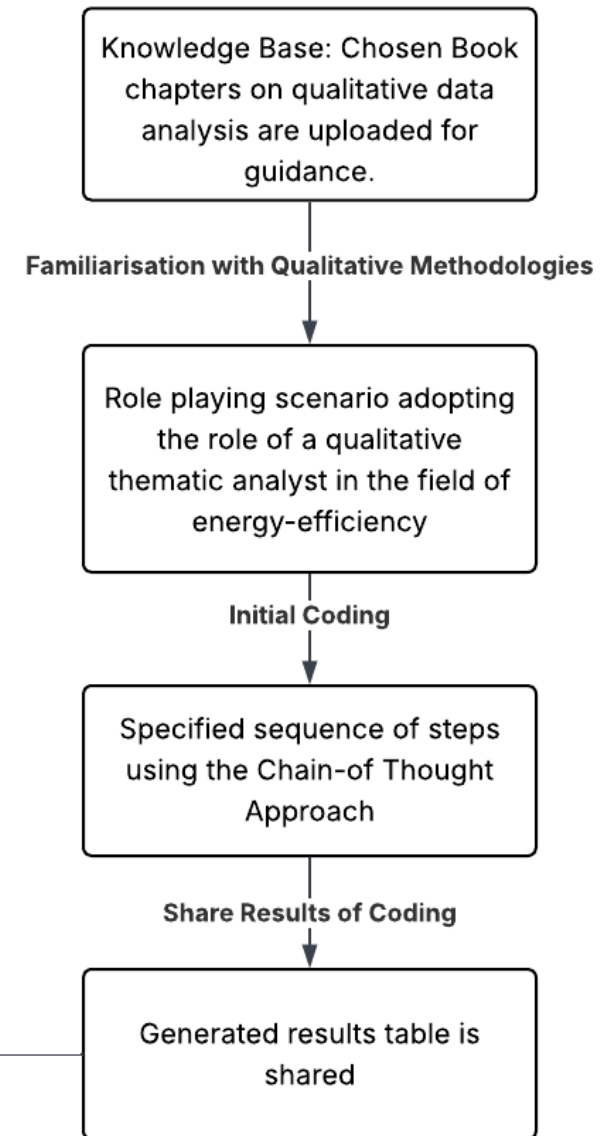
Data Analysis



Manual Thematic Analysis



GPT Model Thematic Analysis



Association for Information Systems

AIS Electronic Library (AISeL)

ICIS 2024 Proceedings

International Conference on Information
Systems (ICIS)

December 2024

Exploring the Potential of Large Language Models (LLMs) for Grounded Theorizing: A Human-in-the-Loop Configuration

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WORKING PAPER

Using ChatGPT for thematic analysis

Authors

Aleksei Turobov, Diane Coyle, Verity Harding

Bennett Institute for Public Policy, University of Cambridge

Date

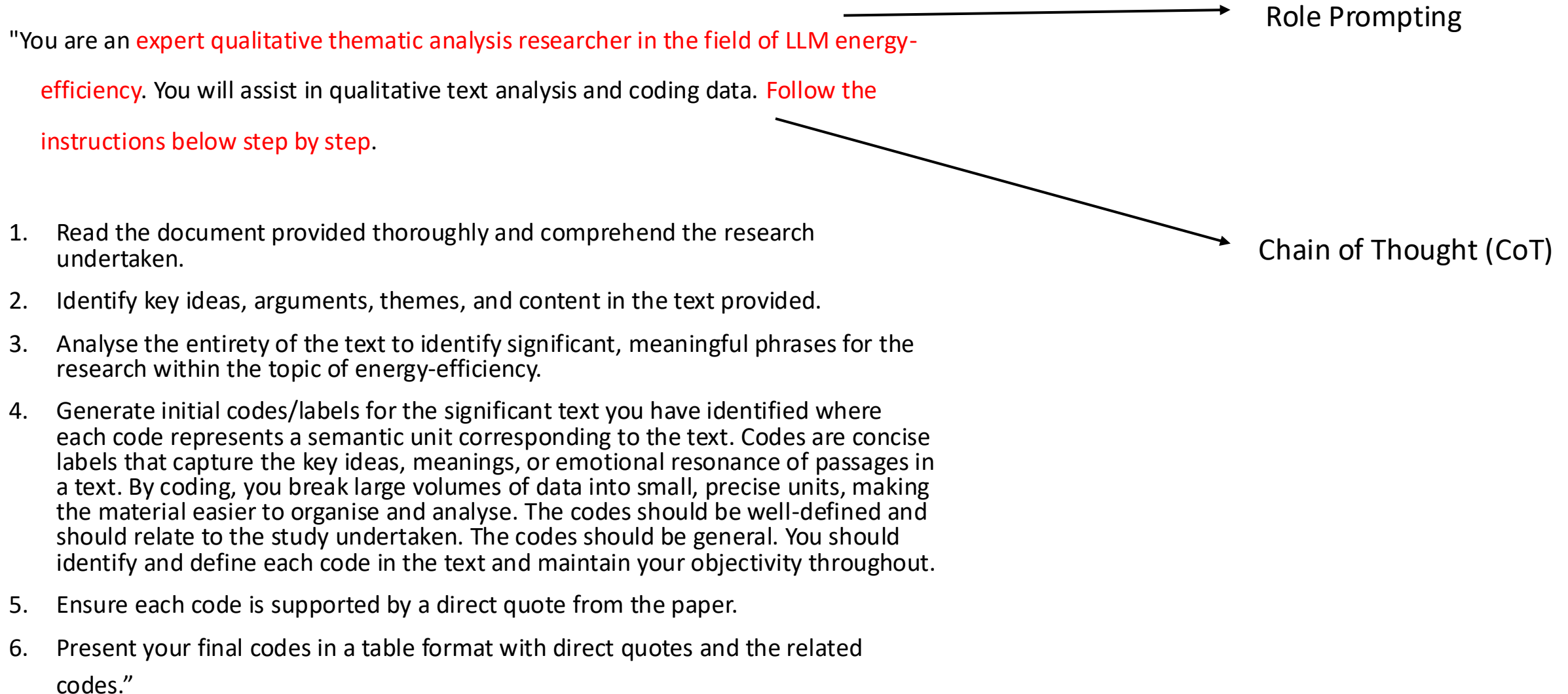
May 2024

Prompt Engineering

"You are an expert qualitative thematic analysis researcher in the field of LLM energy-efficiency. You will assist in qualitative text analysis and coding data. Follow the instructions below step by step.

1. Read the document provided thoroughly and comprehend the research undertaken.
2. Identify key ideas, arguments, themes, and content in the text provided.
3. Analyse the entirety of the text to identify significant, meaningful phrases for the research within the topic of energy-efficiency.
4. Generate initial codes/labels for the significant text you have identified where each code represents a semantic unit corresponding to the text. Codes are concise labels that capture the key ideas, meanings, or emotional resonance of passages in a text. By coding, you break large volumes of data into small, precise units, making the material easier to organise and analyse. The codes should be well-defined and should relate to the study undertaken. The codes should be general. You should identify and define each code in the text and maintain your objectivity throughout.
5. Ensure each code is supported by a direct quote from the paper.
6. Present your final codes in a table format with direct quotes and the related codes."

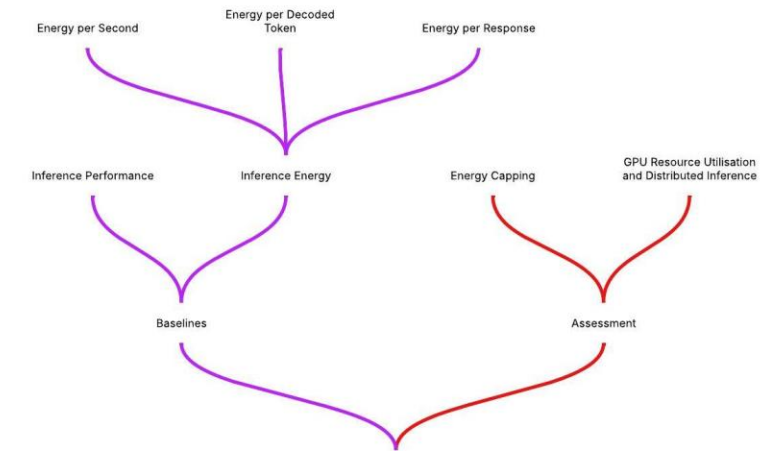
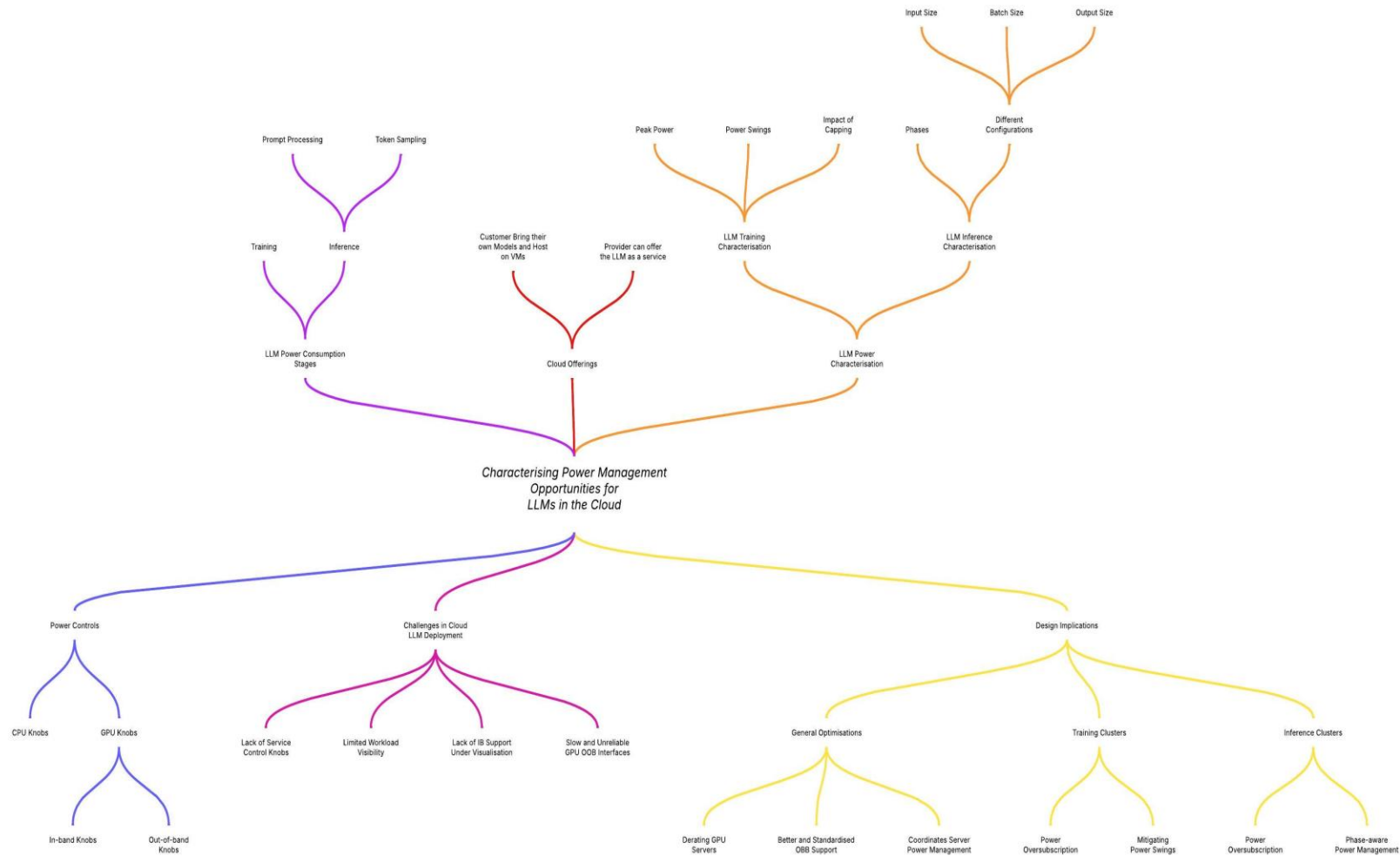
Prompt Engineering



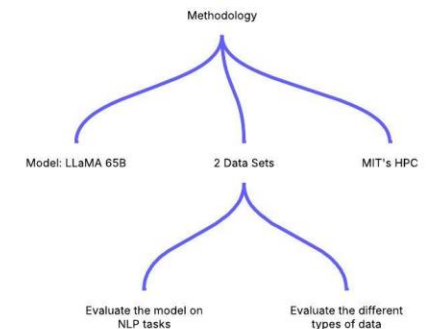
Open Coding

Paper1_Energy	"While these impacts are harder to quantify precisely, we believe that they are nonetheless worth including in the broader environmental impacts of ML."	code 6: Accounting for Energy Use
Paper1_Energy	"we believe it is worth noting the high proportion of energy dedicated to maintaining a LLM like BLOOM in memory (approximately 75% of the total energy consumed by the instance), without it being used."	code 8: Energy Cost of Inference Readiness
Paper2_Energy	"Although training takes months and consumes significant amounts of energy, recent studies show that the energy consumption of LLM serving has now surpassed that of training, leading to significant environmental impacts, especially in terms of carbon footprints."	code 11: Serving Phase Energy Dominance
Paper2_Energy	"serving a single prompt in ChatGPT produces more than 4 grams of CO ₂ eq, which is over 20 times the operational carbon footprint of a web search query."	code 13: High Per-Query Carbon Cost
Paper2_Energy	"Hardware with low embodied carbon may be located in a high carbon intensity region."	code 16: Regional Carbon Intensity Conflict
Paper5_Energy	"Our characterization reveals that LLM training clusters incur massive and coordinated power peaks due to large-scale synchronous training jobs."	code 50: LLM Training Clusters Power Strain
Paper5_Energy	"LLMs deployed in customer operated virtual machines prevent provider visibility into workloads."	code 51: Barriers to Evaluation of Energy Consumption
Paper5_Energy	"Cloud providers and datacenter operators today face a massive GPU capacity crunch due to the explosion in demand for large language models (LLMs)."	code 59: GPU Crunch in Data Centres due to Demand for LLMs

Mind Maps



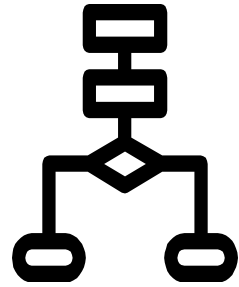
From Words to Watts: Benchmarking the Energy Costs of Large Language Model Inference



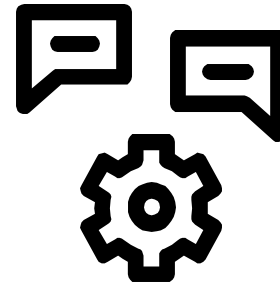
Axial Coding

Quote	Code	Category
<i>"Newer hardware offers better performance but is not always greener due to higher embodied carbon. Older hardware can lower carbon emissions while meeting quality and performance constraints."</i> (Wu, et al., 2025)	"Newer Hardware on Embodied Carbon"	"Embodied Carbon within the Infrastructure"
<i>"SPROUT introduces the concept of "generation directives", a strategy to indirectly manipulate the number of autoregressive inference iterations while providing high-quality content generation."</i> (Li, et al., 2025)	"Inference Carbon Reduction with Directives"	"Algorithm Optimisation and Mitigation Strategies"
<i>"simple interventions like GPU power capping is available to be deployed today - our preliminary analysis with LLM inference in this paper suggests that power capping can be an effective tool for reducing inference energy."</i> (Samsi, et al., 2023)	"Power Capping for Mitigating High Energy Use"	"Operational Efficiency and Energy-Sensitive Scheduling"
<i>"Advanced hardware offers higher performance but is not always greener due to higher embodied carbon"</i> (Wu, et al., 2025)	"Performance vs Sustainability Trade-Off"	"Energy Trade-offs and Infrastructure Constraints"
<i>"sustainability is not, at its root, a technical problem, nor will merely technical solutions be sufficient. Instead, deep economic, political, and cultural adjustments will ultimately be required, along with a major, long-term commitment in each sphere to deploy the requisite technical solutions at scale."</i> (Ren, et al., 2024)	"Socio-Technical Sustainability Challenge"	"Broader Environmental and Societal Impacts"
<i>"The AI boom, driven by the demand for generative artificial intelligence (GenAI) has prompted concerns over its environmental impact, particularly in terms of carbon emissions associated with the datacenters hosting these technologies."</i> (Li, et al., 2025)	"Carbon Emission Concerns Regarding Data Centres"	"Operational Carbon & Resource Demands"
<i>"LLMs deployed in customer operated virtual machines prevent provider visibility into workloads."</i> (Patel, et al., 2024)	"Barriers to Evaluation of Energy Consumption"	"Evaluation Metrics and Benchmarking for Sustainability"
<i>"Our measurement on a single Nvidia A100 GPU over 5,000 testing prompts without optimal batching showed that Gemma-2B-it consumes about 0.0002 kWh for producing a 500-word response, roughly 1% energy of Llama-3-70B for doing the same job."</i> (Ren, et al., 2024)	"Carbon-Impact and Size Correlation"	"Model Architecture and Algorithm Design"

Selective Coding



Deciding on a
core theme



Identifying
emerging
relationships

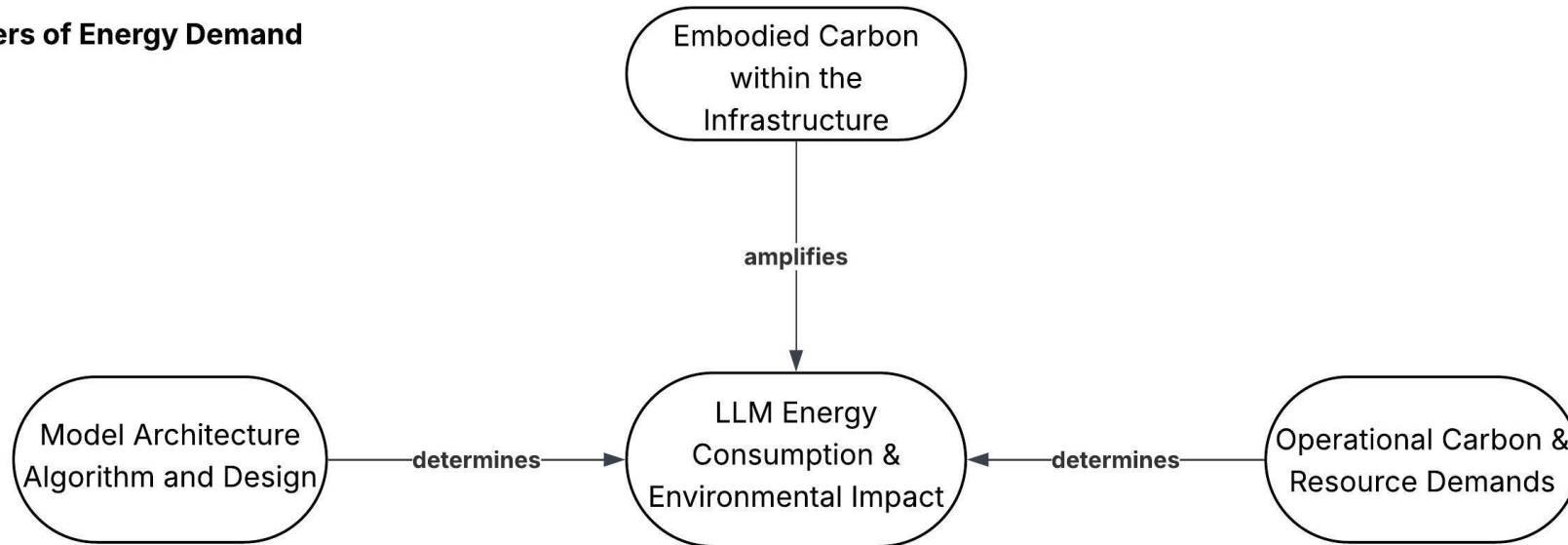
03

Findings



Drivers of Energy Demand

Drivers of Energy Demand



Embodied Carbon within the Infrastructure

Quote

"The embodied emissions are those emissions associated with the materials and processes involved in producing a given product, such as the computing equipment needed to train and deploy ML models."

"Embodied carbon represents the carbon emissions associated with the manufacturing and packaging of computer components, effectively "embodied" within the device itself."

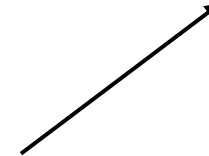
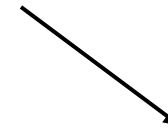
Open Code

"code 5: Embodied Carbon Emissions"

Axial Code

Embodied Carbon
within the Infrastructure

"code 48: Embodied Carbon Emissions within the Device"



Operational Carbon and Resource Demands

Quote

"Cloud providers and datacenter operators today face a massive GPU capacity crunch due to the explosion in demand for large language models (LLMs)."

"Power, space, and cooling are the major bottlenecks in datacenter provisioning. However, for GPU-heavy workloads like LLMs, power is the main bottleneck."

"serving a single prompt in ChatGPT produces more than 4 grams of CO₂eq, which is over 20 times the operational carbon footprint of a web search query."

Open Code

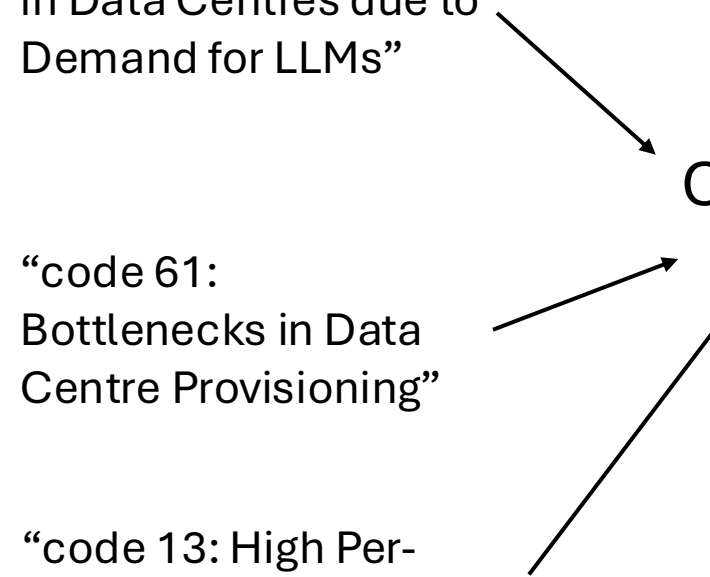
"code 59: GPU Crunch in Data Centres due to Demand for LLMs"

"code 61: Bottlenecks in Data Centre Provisioning"

"code 13: High Per-Query Carbon Cost"

Axial Code

Operational Carbon and Resource Demands



Model Architecture and Algorithm Design

Quote

"We first observe that under the same configuration, larger models (e.g., BLOOM-176B) incur a much higher amount of computation during both prompt and token phases and show significantly larger peak and mean power consumption."

"Current LLMs are stacked by transformer blocks, each containing a Multi-Head Attention (MHA) layer, LayerNorms, and a Feed-Forward Network (FFN)."

Open Code

"code 60: Larger Model's Impact on Energy Demand"

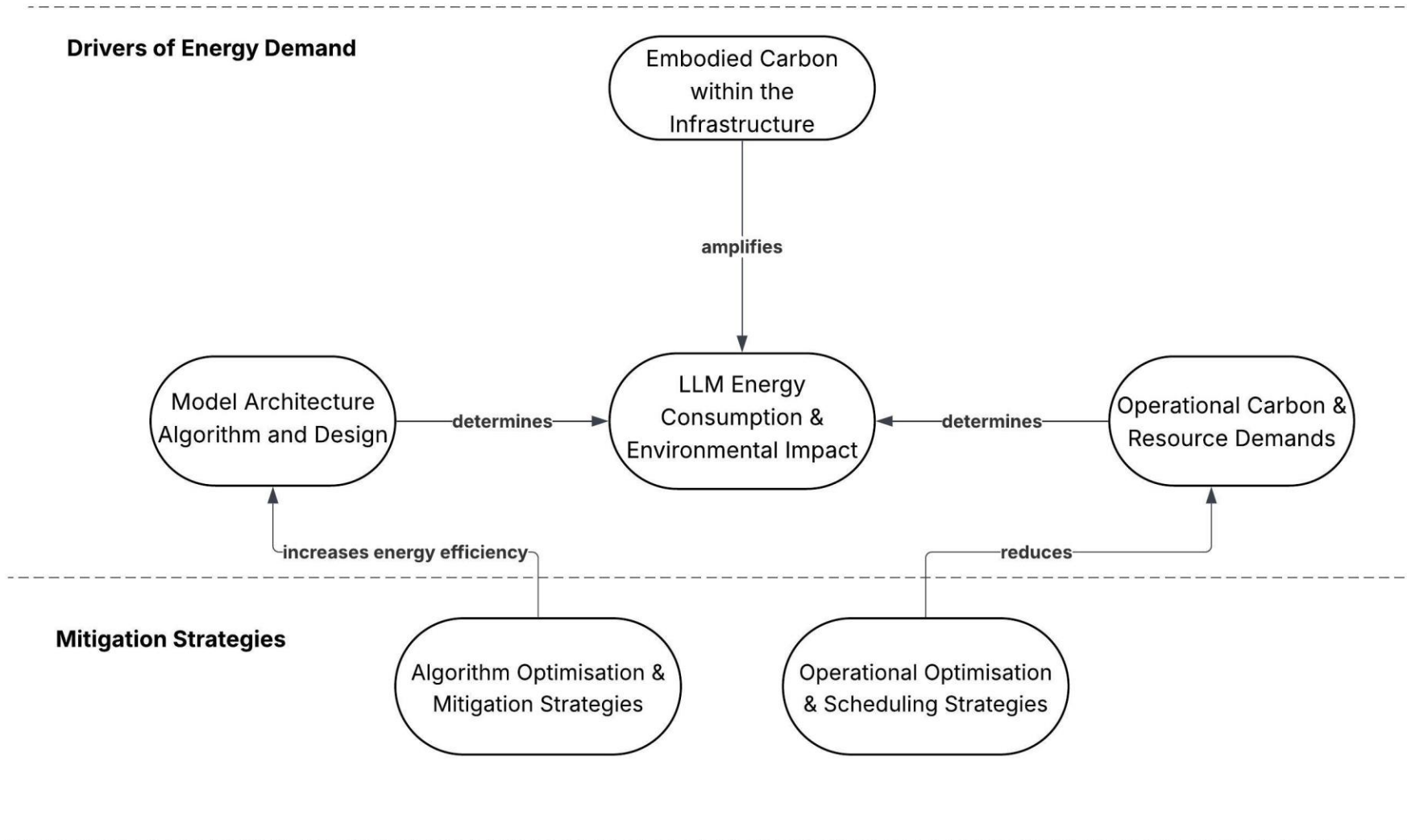
"code 107: Transformer Architecture"

Axial Code

Model Architecture and Algorithm Design

```
graph LR; Q1["We first observe that under the same configuration, larger models (e.g., BLOOM-176B) incur a much higher amount of computation during both prompt and token phases and show significantly larger peak and mean power consumption."] --> MACAD["Model Architecture and Algorithm Design"]; Q2["Current LLMs are stacked by transformer blocks, each containing a Multi-Head Attention (MHA) layer, LayerNorms, and a Feed-Forward Network (FFN)."] --> MACAD; OC1["code 60: Larger Model's Impact on Energy Demand"] --> MACAD; OC2["code 107: Transformer Architecture"] --> MACAD;
```

Mitigation Strategies



Algorithm Optimisation & Mitigation Strategies

Quote

"Model quantization reduces model sizes and power usage, thereby enabling more workloads to be deployed under a power budget."

"SPROUT introduces "generation directives" to guide the autoregressive generation process, achieving a balance between ecological sustainability and high-quality outputs."

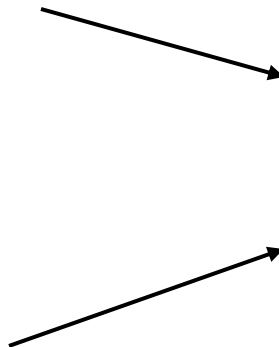
Open Code

"code 58: Quantisation-Driven Energy Saving"

"code 37: Carbon-Efficient Generation Control"

Axial Code

Algorithm
Optimisation &
Mitigation Strategies



```
graph LR; C58["code 58: Quantisation-Driven Energy Saving"] --> AOS["Algorithm Optimisation & Mitigation Strategies"]; C37["code 37: Carbon-Efficient Generation Control"] --> AOS;
```

Operational Efficiency and Energy Sensitive Scheduling

Quote

"Our deductions align with data from production LLM clusters, revealing that inference workloads offer substantial headroom for power oversubscription."

"Many LLMs have parameter sizes exceeding the capacity of a single GPU. Therefore, it is necessary to partition them across distributed GPUs and execute them in a model parallel fashion."

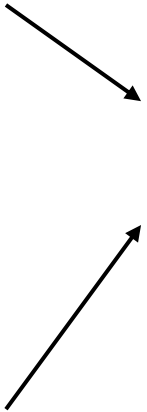
Open Code

"code 49: Inference Power Flexibility"

"code 32: Parallel Execution for Superior Performance"

Axial Code

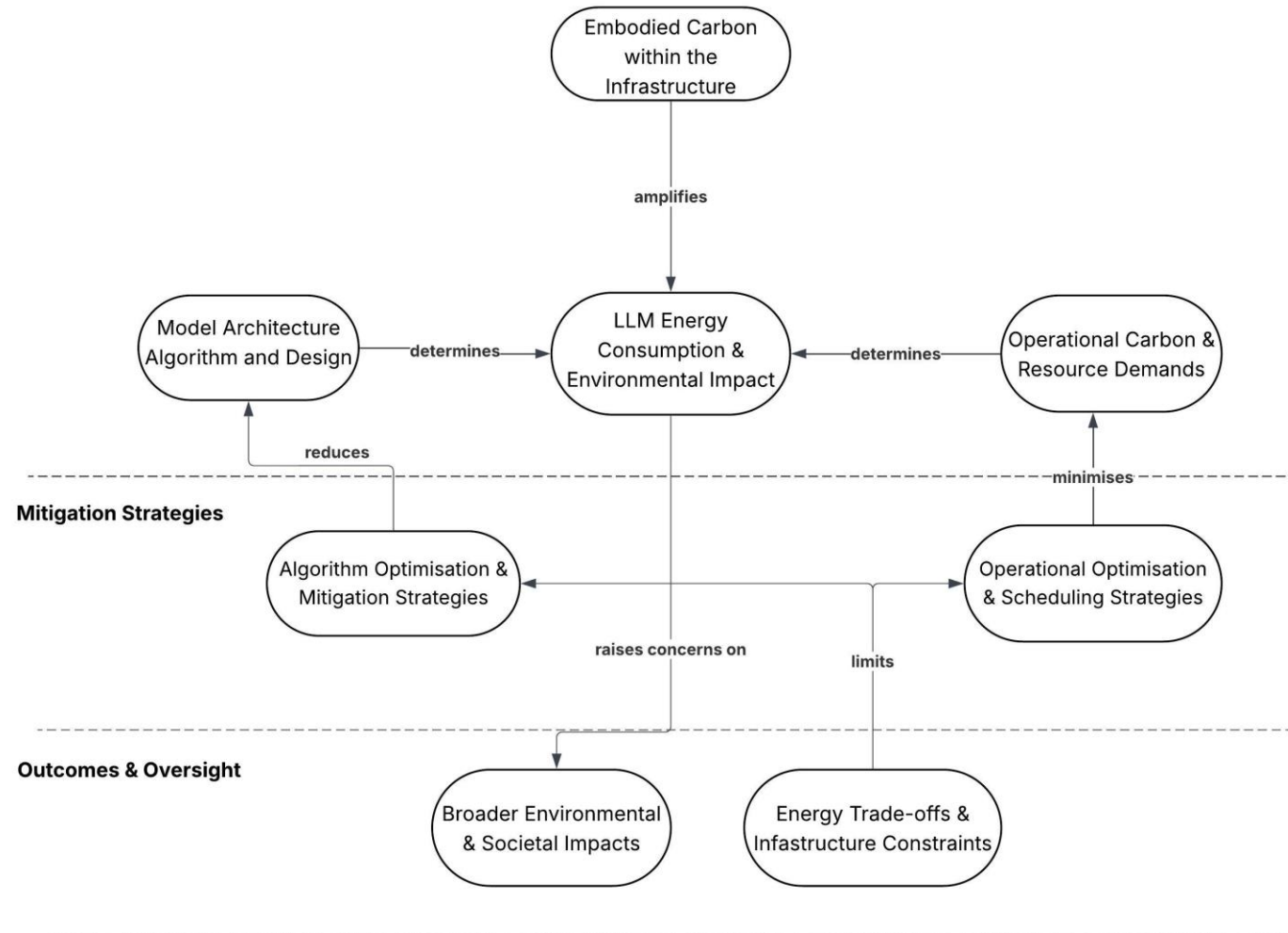
Operational Efficiency and Energy Sensitive Scheduling



```
graph LR; C49["code 49: Inference Power Flexibility"] --> Title["Operational Efficiency and Energy Sensitive Scheduling"]; C32["code 32: Parallel Execution for Superior Performance"] --> Title;
```

Outcomes & Oversight

Drivers of Energy Demand



Broader Environmental & Societal Impacts

Quote

“the growing size of LLMs may substantially increase their energy consumption and lower the human-to-LLM ratios, highlighting the need for further research to ensure the sustainability and efficiency of LLMs.”

“sustainability is not, at its root, a technical problem, nor will merely technical solutions be sufficient. Instead, deep economic, political, and cultural adjustments will ultimately be required, along with a major, long-term commitment in each sphere to deploy the requisite technical solutions at scale.”

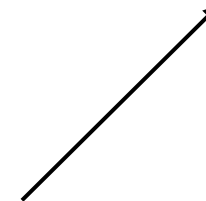
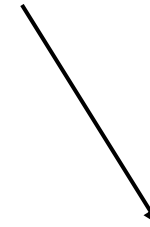
Open Code

“code 78: Relative Environmental Efficiency”

“code 87: Socio-Technical Sustainability Challenge”

Axial Code

Broader Environmental & Societal Impacts



Energy Trade-offs and Infrastructure Constraints

Quote

"Minimizing the total carbon footprint of LLM serving involves several uncertainties, including inaccurate performance and energy profiling, unreliable performance and energy modelling, uncertain embodied carbon accounting, unpredictable carbon intensity, and inconsistent communications between servers."

"This pattern reaffirms that choosing the optimal model and hardware combination requires a balance between performance needs and carbon efficiency."

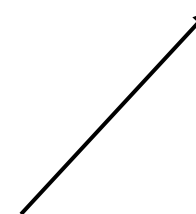
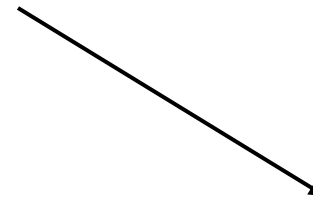
Open Code

"code 17: Carbon Minimisation Uncertainties"

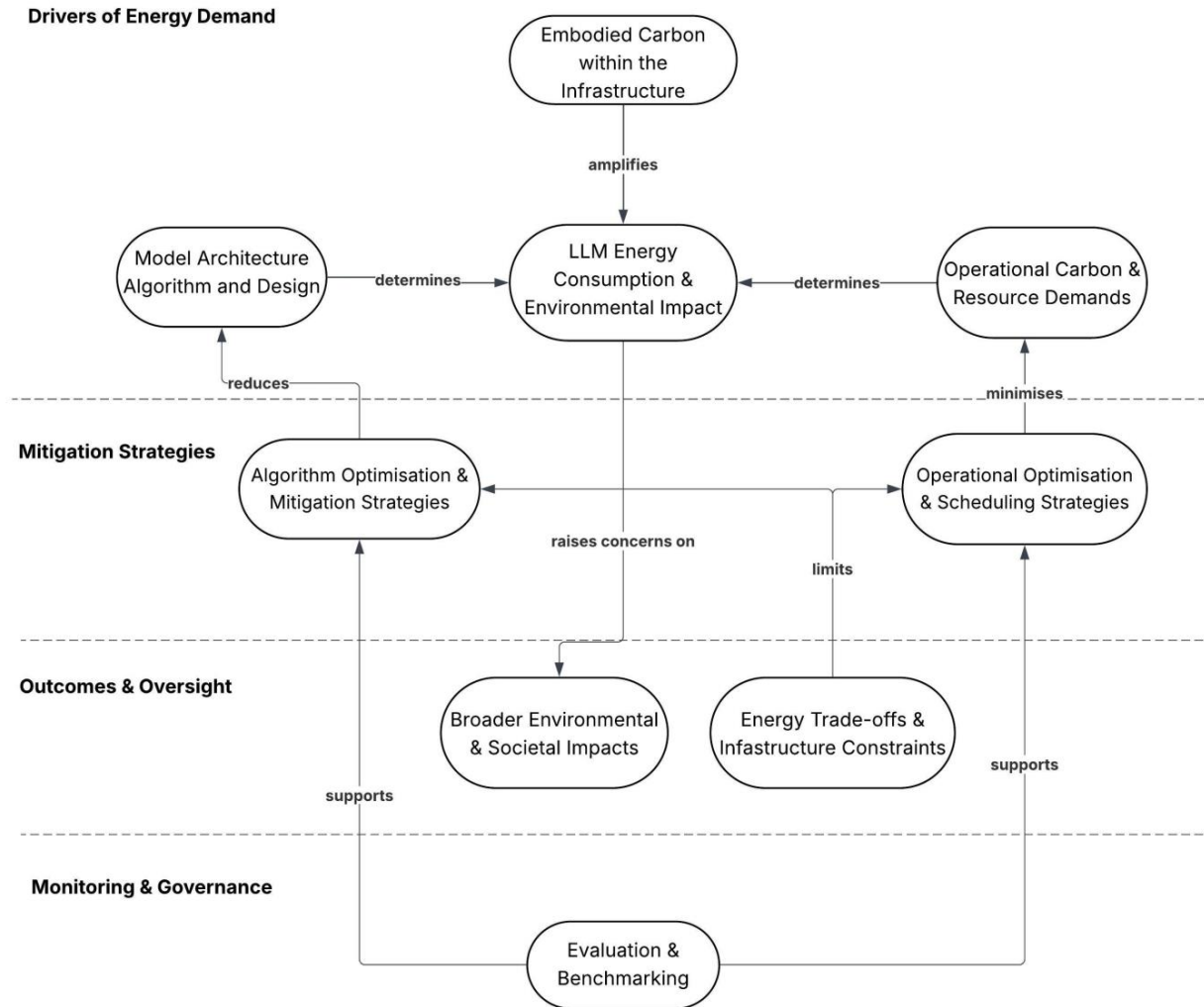
Axial Code

Energy Trade-offs and Infrastructure Constraints

"code 75: Holistic Deployment Optimisation"



Monitoring and Governance



Evaluation & Benchmarking

Quote

“we emphasize the importance of continuing and strengthening research efforts to ensure the long-term sustainability of LLMs.”

"it is crucial to build a generalized benchmarking framework that bridges different LLM optimization techniques with various models and libraries, facilitating seamless integration and evaluation."

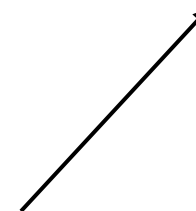
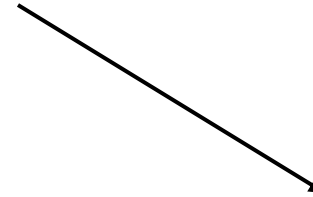
Open Code

“code 80: Call for Continuous Research”

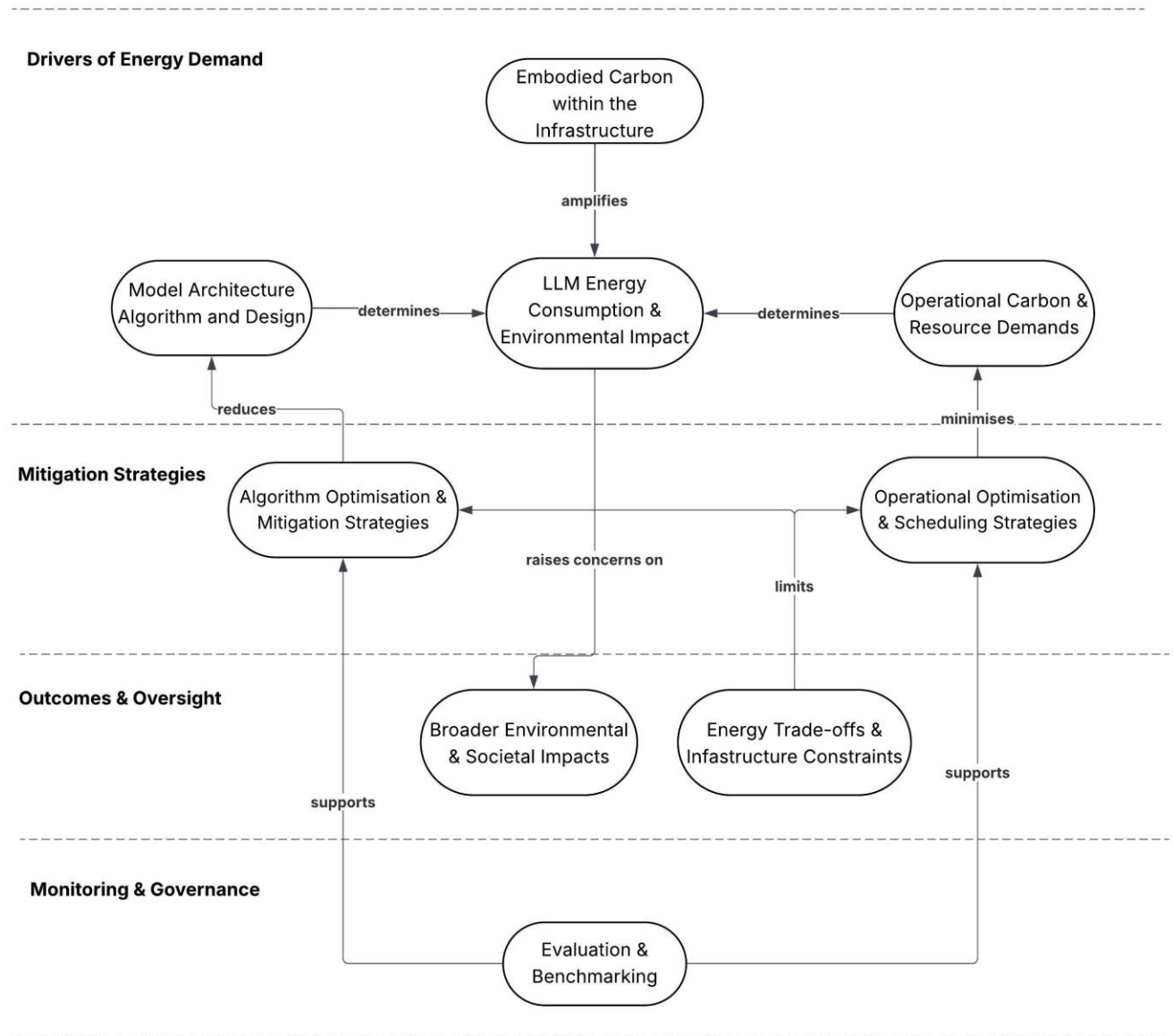
“code 18: Unified Sustainability Benchmarking”

Axial Code

Evaluation and Benchmarking



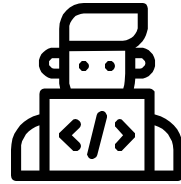
Integrative Conceptual Framework



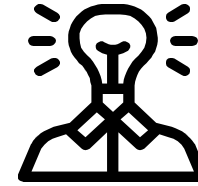
Conclusion – Key Findings



Deployment Flexibility



Community-
driven Efficiency
Initiatives



Societal
Benchmarks

Implications of Findings



Researchers



Developers



Recommendations

Any Questions?